

well labelled. It means that some of the data present in the data sets are already tagged with correct answers. Thereafter, the same computer or machine is provided with a new set of data that applies the supervised learning algorithm, analyses the data and then produces a correct outcome from the labelled data.

For example, if we are given a basket filled with different varieties of fruits, when implementing supervised learning, the first step will be to train the machine with respect to the different fruits, one by one. This involves teaching the machine to recognise the different categories so that this knowledge can be applied to the rest of the fruits.

Supervised learning is categorised into two types of algorithms.

Classification: A classification algorithm is used when the output variable expected is in terms of the name of a category, like red, blue, etc.

Regression: A regression algorithm is applied when the output variable expected is in terms of any real value, such as dollars, weight, etc.

Unsupervised machine learning: This is the training of machines using the available set of information that is neither categorised nor labelled. It directly allows the algorithm to act on the information set without any guidance. In unsupervised ML, it will be the task of the machine to group unsorted data according to patterns, similarities and differences, without any prior training. Unlike in the case of supervised learning, no such training is provided to the machine. Instead, the machine needs to find the hidden structure in the given set of unlabelled data on its own.

For example, let's consider an image that has both dogs and cats, but no information or label is provided with respect to these animals. In this case, unsupervised learning needs to be applied to help machines categorise them.

Unsupervised learning is categorised into the following two types of algorithms.

Clustering: A clustering algorithm is one that is applied wherever we want to discover the inherent groupings in the available data sets, like grouping customers on the basis of their purchasing behaviour.

Association: An association rule algorithm is applied wherever we need to discover a set of rules that describes a large portion of the data, like identifying that the people who buy X are likely to buy Y.

Reinforcement machine learning: This third category focuses on behavioural psychology. It looks at how software agents should take specific action in any environment to maximise some notion, in order to achieve the cumulative reward. The rewards can be winning a game or earning more money.

Python — an open source programming language for machine learning

Python is simply the Swiss army knife of ML. It is one of the open source programming languages widely used to perform complex operations and has a full suite of tools for increasing the productivity of ML.

Python is popular among its users because it is easy to learn and pretty simple when it comes to programming. It is considered to be one of the most consistent math-like programming languages. Python has its own set of built-in functions and utilities, which help it perform a plethora of complex operations with just a few lines of code. It is also way ahead in terms of its easy syntactical character when compared to other programming languages.

Python has a large set of libraries that can be easily used for machine learning, such as SciPy, NumPy, ScikitLearn, PyBrain, etc. It can be used to develop code in the Map-Reduce model even while working in the Hadoop ecosystem. Spark, one of the modern technologies used for scalable Big Data analysis, has also got its machine learning libraries written in Python. Simplicity and wide applicability make Python a popular ML language.

At times, the code developed in Python almost seems to be written in the English language. The way Python mirrors human language or its mathematical counterparts makes ML a bit easier. As per the TIOBE programming index, Python is ranked No. 5, which is way above other programming languages that are used for ML and data analysis, like R. Python has outperformed R in the fields of data science and ML for the last five years.

Python's capabilities are given a big boost by the different available packages in ML and data analytics. Pandas, one of the best known data analysis packages, gives Python high-performance structures and several data analysis tools as well. Python excels in offering a playground for playing with data, not just numerically, but also for the following other functions:

- Downloading varied contents from websites and APIs
- Interfacing with different databases and spreadsheets
- Manipulating audio, text and images
- Availability of sophisticated tools for the exploration and presentation of results, like Pandas, Jupyter, etc.

Why is Python suitable for machine learning?

Here are some important reasons why Python is considered ideal for ML.

1. Python is an easy to learn and developer-friendly programming language that is pretty simple to start with. Its syntax is simple and the code can be written in fewer lines as compared to other programming languages.
2. It has numerous built-in packages for ML and other computations — for example, Numpy, Keras, Pandas, etc. All these packages are well-documented, and hence are quite helpful in starting with any project or solution. It accelerates the process of fixing bugs.
3. All the available libraries of Python are quite powerful. They comprise many features that are helpful in performing complex computations. These help in fast, efficient and stable development. Python also uses a wide range of computation speed improvements continuously to improve the performance of libraries.